

Bloom Filter-Based MPSI

Weekly Progress Meeting 16 April 2026

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Bibliography

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- [2] Ou Ruan, Changwang Yan, Jing Zhou, and Chaohao Ai. A practical multiparty private set intersection protocol based on bloom filters for unbalanced scenarios. *Applied Sciences*, 13(24):13215, 2023.
- [3] Florian Kerschbaum. Outsourced private set intersection using homomorphic encryption. In *Proceedings of the 7th ACM Symposium on Information, Computer and Communications Security*, pages 85–86, 2012.
- [4] Falzon, Francesca, and Evangelia Anna Markatou. "Re-visiting authorized private set intersection: A new privacy-preserving variant and two protocols." *Proceedings on Privacy Enhancing Technologies* (2025).
- [5] Koster, Tjitske Ollie, Francesca Falzon, and Evangelia Anna Markatou. "Bandwidth Efficient Partial Authorized PSI." *Cryptology ePrint Archive* (2025).
- [6] Lindell, Yehuda. "How to simulate it—a tutorial on the simulation proof technique." *Tutorials on the Foundations of Cryptography: Dedicated to Oded Goldreich* (2017): 277-346.

Progress

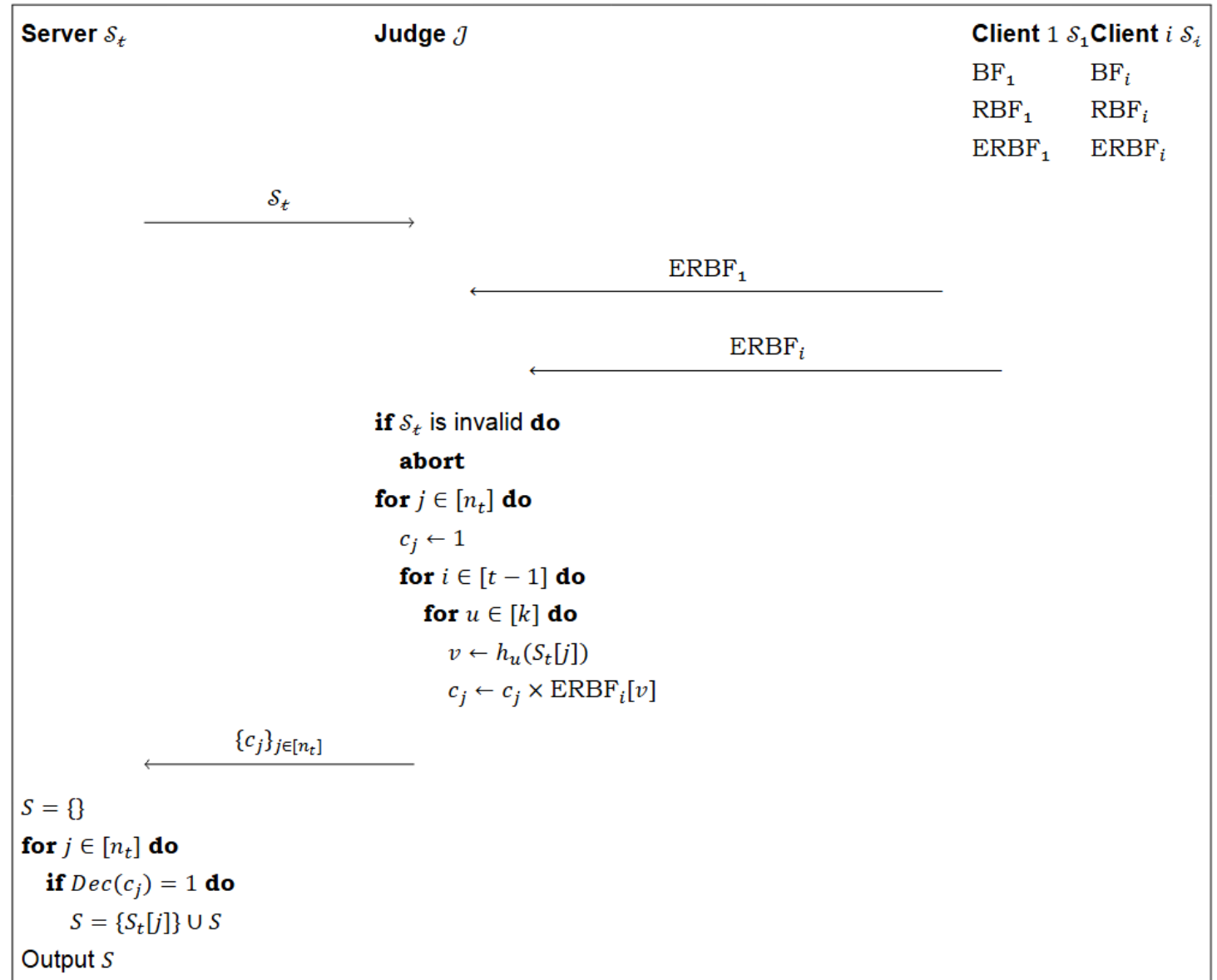
- Brainstorm on APSI1/PAPSI2
- Writing!

Progress

APSI 2 – Active Judge

Threat Model:

- Malicious Server
- Semi-honest Judge
- Semi-honest Clients



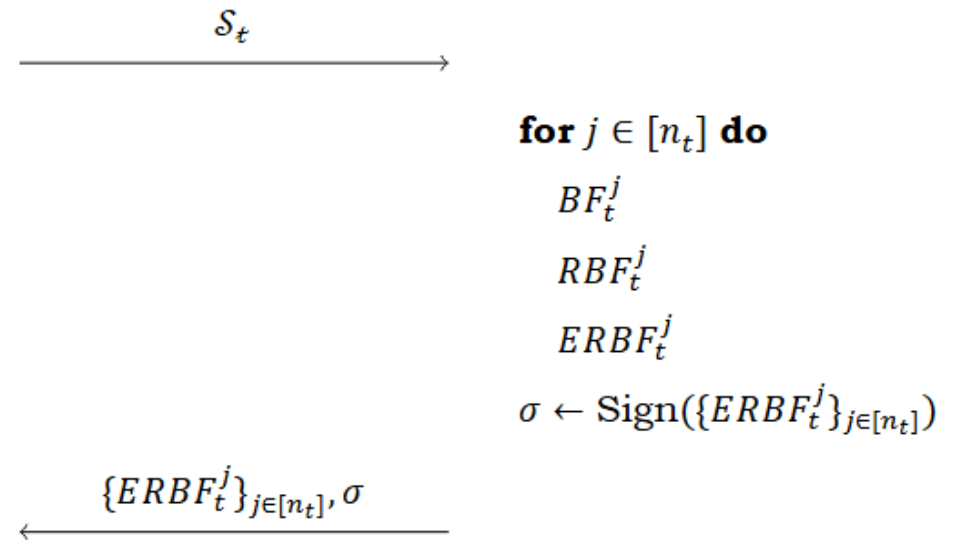
Progress

APSI 3 – Tired Judge

- VERY large bandwidth ☹️
- Server knows how many bins were set for each element, but not which bins
- Clients have to jointly decrypt, since if the server has the decryption key, they can decrypt their own ERBF after the authorization phase and know which bin corresponds to which element
- Hash functions are not allowed to give the same bin for the same element, otherwise the protocol will have false negatives

Authorize
Server $\mathcal{S}_t$

Judge $\mathcal{J}$



Progress

APSI 3 – Tired Judge

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- Hash functions are not allowed to give the same bin for the same element, otherwise the protocol will have false negatives

Intersect
Server S_t

Client 1 S_1

BF_1
 RBF_1
 $ERBF_1$

...

Client i S_i

BF_i
 RBF_i
 $ERBF_i$

$ERBF_2$

$ERBF_i$

$\{ERBF_t^j\}_{j \in [n_t]}, \sigma$

VerifySignature($\{ERBF_t^j\}_{j \in [n_t]}, \sigma$)

for $j \in [n_t]$ **do**

for $v \in [m]$ **do**

$ERBF_{agg}^j[v] \leftarrow ERBF_t^j[v] \times \prod_{i=1}^{t-1} ERBF_i[v]$

ShuffleAndRerandomize($ERBF_{agg}^j$)

$\{ERBF_{agg}^j\}_{j \in [n_t]}$

$\{ERBF_{agg}^j\}_{j \in [n_t]}$

for $j \in [n_t]$ **do**

for $v \in [m]$ **do**

$ERBF_{sh_i}^j[v] \leftarrow (ERBF_{agg}^j[v])^{d_i \times sk_i}$

$\{ERBF_{sh_2}^j\}_{j \in [n_t]}$

$\{ERBF_{sh_i}^j\}_{j \in [n_t]}$

$S = \emptyset$

for $j \in [n_t]$ **do**

$BF_{agg}^j \leftarrow \text{CombDecrShares}(\{ERBF_{sh_i}^j\}_{i \in [t-1]})$

if CountBins(BF_{agg}^j) = k **do**

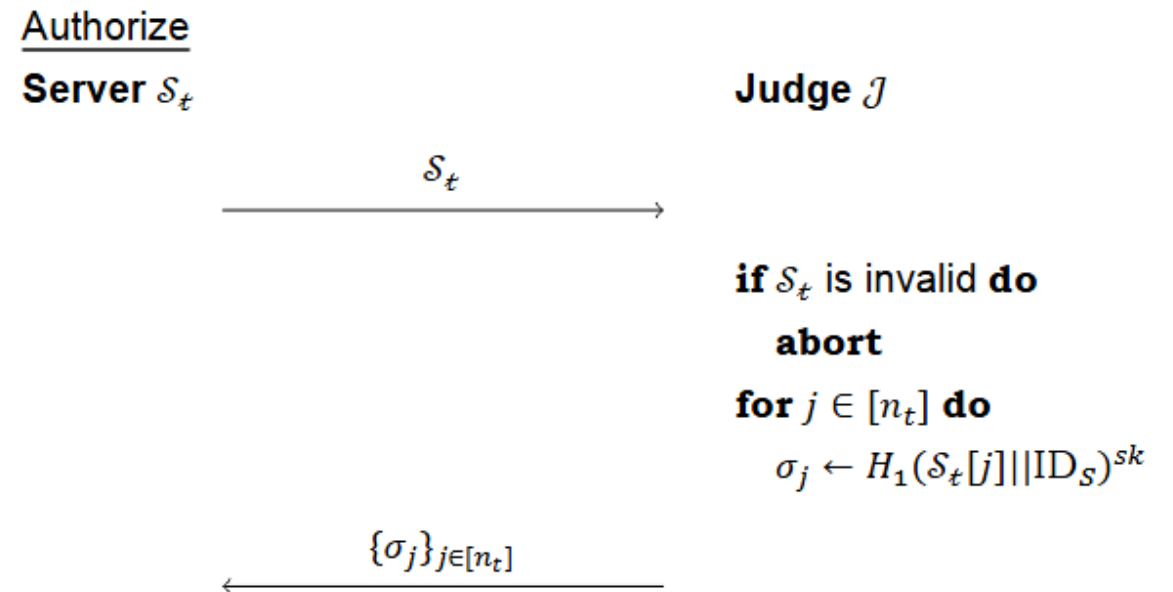
$S = \{S_t[j]\} \cup S$

Output S

Progress

APSI 4 – Lilika’s Judge

- There can be collisions in H_1 (which add to the FP probability), but they are negligible
- The server has the secret (decryption) key, so they can decrypt the combined *ERBF*
 - This should not be a problem, since they don’t have s
 - If it is a problem, we can do Shamir secret sharing for decryption instead (adds one round of interaction)
- Do we even need encryption?



Progress

APSI 4 – Lilika’s Judge

- There can be collisions in H_1 (which add to the FP probability), but they are negligible
- The server has the secret (decryption) key, so they can decrypt the combined $ERBF$
 - However, they don’t have s
 - If it is a problem, we can do Shamir secret sharing for decryption instead (adds one round of interaction) – augmented semi-honest server
 - Or we could do OT
 - GBF

Intersect
Server S_t

Client 1 S_1

$s \leftarrow \mathbb{Z}_q$

...

Client i S_i

s

s

for $j \in [n_1]$ **do**

$\hat{y}_{1,j} \leftarrow e(H_1(S_1[j] || ID_S), pk^S)$

$BF_1 \leftarrow \text{ComputeBF}(\{\hat{y}_{1,j}\}_{j \in [n_1]})$

RBF_1

$ERBF_1$

$ERBF_2$

$ERBF_i$

$S \leftarrow g_2^S$

for $v \in [m]$ **do**

$ERBF_{\text{agg}}[v] \leftarrow \prod_{i=1}^{t-1} ERBF_i[v]$

$s, ERBF_{\text{agg}}$

$I = \emptyset$

for $j \in [n_t]$ **do**

$\hat{x}_j \leftarrow e(\sigma_j, S)$

$c_j \leftarrow 1$

for $u \in [k]$ **do**

$c_j \leftarrow c_j \times ERBF_{\text{agg}}[h_u(\hat{x}_j)]$

if $\text{Dec}(c_j) = 1$ **do**

$I = \{S_t[j]\} \cup I$

Output I

for $j \in [n_i]$ **do**

$\hat{y}_{i,j} \leftarrow e(H_1(S_i[j] || ID_S), pk^S)$

$BF_i \leftarrow \text{ComputeBF}(\{\hat{y}_{i,j}\}_{j \in [n_i]})$

RBF_i

$ERBF_i$

Next Week

- Start implementing APSI2+PAPSI2 / APSI3 / APSI4+PAPSI4
- Writing!

Thank you!

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